

Advanced Machine Learning

Lab 1 --- Module 1

Instructions on how to submit your work on Canvas

To convert your Notebook to an HTML file, follow these steps:

- Open the Notebook that you want to convert to an HTML file.
- Ensure that the notebook is saved with the latest changes.
- In the Notebook interface, navigate to File in the menu bar.
- Click on Download as and select HTML (.html) from the dropdown menu. This will initiate the conversion process.
- The Jupyter Notebook will be converted to an HTML file and automatically downloaded to the default download location on your computer.
- Locate the downloaded HTML file on your computer and verify that the conversion was successful.
- Optionally, open the HTML file in a web browser to view and verify the content.
- Upload your HTML file on Canvas.

Make sure to save your Notebook before attempting to convert it to an HTML file.

Exercise 1 (1 point)

MNIST dataset, is a set of 70,000 small images of digits handwritten by high school students and employees of the US Census Bureau. Each image is labeled with the digit it represents. Create the data matrix $\mathbf{X}$ as well as the target variable $\mathbf{y}$. Check and display the dimensions of $\mathbf{X}$ and $\mathbf{y}$. Display some of the digits in the dataset. Then identify one digit for example, the number 5. The 5-detector will be your binary classifier, capable of distinguishing between just two classes, 5 and not 5.

Hint: For the first task use the code that is provided to you through notebook

`SampleCode_Lab1` .

Exercise 2 (2 points)

Prepare a training dataset by selecting the first 60,000 observations and create a testing dataset by selecting the last 10,000 images. Make sure that you shuffle the indices of your

training observations and their class labels. For this task pick the Stochastic gradient descent classifier from the Scikit-Learn's `SGDClassifier` class and train it to distinguish the two classes.

Exercise 3 (2 points)

Evaluate the performance of your classifier by

1. Measuring accuracy using the function `cross_val_score()` from `Scikit-Learn`. In addition, implement cross-validation yourself using the `StratifiedKFold` class. Check if your implementation produces the similar results with the results obtained using `cross_val_score()`.
2. The use of the confusion matrix.
3. Understanding the precision/recall trade-off. In addition, print the precision score, recall score, and f1 score.
4. The use of the ROC curve.
5. Compare the ROC curve generated by the `RandomForestClassifier` with the ROC curve generated by the `SGDClassifier`. Which method performs better?